

Total No. of Questions : 6]

SEAT No. :

P18

[Total No. of Pages : 2

APR-17/BE/Insem.-19
B.E. (Mechanical)
ROBOTICS
(2012 Pattern) (Elective - III) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Figures to the right indicate full marks.*
- 2) *Draw neat figures wherever necessary.*
- 3) *Use of scientific calculators is allowed.*

- Q1) a)** State three laws of Robotics. **[6]**
b) Explain sensor selection criteria for robotic applications. **[4]**

OR

- Q2) a)** Classify Robots and explain any one Robot configuration in detail. **[6]**
b) Suggest which configuration robot is suitable for following application and justify. **[4]**
i) Spray Painting
ii) Pick & Place

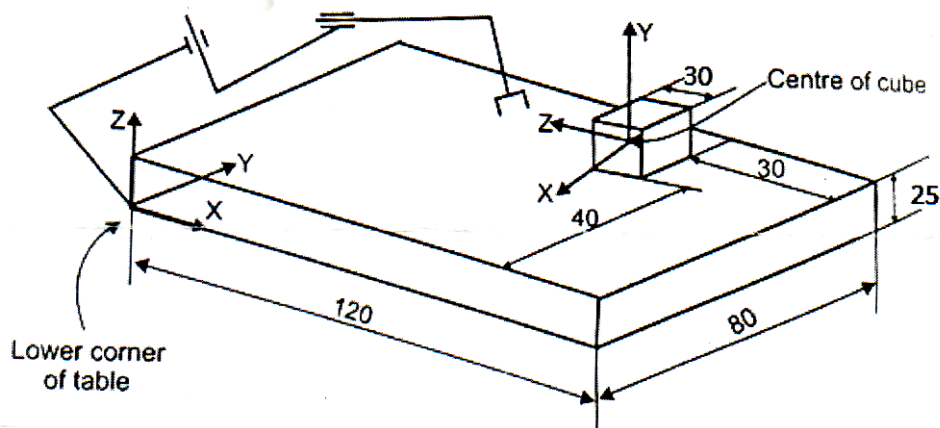
- Q3) a)** Explain with neat sketch derivation of link transformation. **[6]**
b) What is meant by Forward position kinematics of a robot. **[4]**

OR

- Q4) a)** Explain the steps involved in DH notation process. **[6]**
b) A robot shown in following fig. is used to grip a cubical object so that the gripper moves towards centre of the object in the opposite direction to that of X-axis of the object. The fingers of the gripper move parallel to X-axis of robot base. Determine orientation of gripper with respect to the base co-ordinate frame of the robot and also with respect to the

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object's co-ordinate frame. Also determine orientation and position of object with respect to the robot. [4]



Q5) Explain singular continuity for planar 4-bar mechanism. [10]

OR

Q6) a) What is the physical significance of angular velocity vector. [4]

b) Write a short note on [6]

i) Jacobian

ii) Singularities.

